

Differential Calculus — Chain Rule

Chapter 2: Power of Tangent Function | Page 33

PROBLEM STATEMENT

Tangent Square Derivative

Find the derivative with respect to x of the function:

$$y = 3 \tan^2 x$$

STEP 1 প্রদত্ত ত্রিকোণমিতিক ঘাত ফাংশনটি উল্লেখ করি (Given Function)

$$y = 3 \tan^2 x = 3(\tan x)^2$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} [3(\tan x)^2] = 3 \cdot \frac{d}{dx} [(\tan x)^2]$$

STEP 3 শক্তি বিধি ও চেইন রুল প্রয়োগ করি $(\frac{d}{dx}[u^2] = 2u \frac{du}{dx})$

$$\frac{dy}{dx} = 3 \times 2(\tan x)^{2-1} \cdot \frac{d}{dx}(\tan x) = 6 \tan x \cdot \frac{d}{dx}(\tan x)$$

STEP 4 অভ্যন্তরীণ ট্যানজেন্ট ফাংশনের অন্তরক নির্ণয় করি (Differentiating Inner Tangent Function)

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

STEP 5 মান বসিয়ে চূড়ান্ত অন্তরজ পাই (Final Product Form)

$$\frac{dy}{dx} = 6 \tan x \sec^2 x$$

FINAL ANSWER
$$\frac{dy}{dx} = 6 \tan x \sec^2 x$$

Rigorous Insights & Mark Scheme

AMIE Section-A Standard Evaluation Protocol | Page 33, HW 07

BOX 1: FORMULATION & TANGENT POWER RULES

- **Power Chain Formula:** $\frac{d}{dx}[c \tan^n x] = cn \tan^{n-1} x \sec^2 x$.
- For $c = 3$ and $n = 2$:

$$\frac{dy}{dx} = 3 \times 2 \tan^{2-1} x \sec^2 x = 6 \tan x \sec^2 x$$

- **Alternative Sine-Cosine Form:**

$$6 \tan x \sec^2 x = 6 \left(\frac{\sin x}{\cos x} \right) \left(\frac{1}{\cos^2 x} \right) = \frac{6 \sin x}{\cos^3 x}$$

BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Confusing $\tan^2 x$ with $\tan(x^2)$:** $\frac{d}{dx}(\tan^2 x) = 2 \tan x \sec^2 x$, whereas $\frac{d}{dx} \tan(x^2) = 2x \sec^2(x^2)$.
- **Forgetting the Constant 3:** Multiplying $2 \tan x$ without multiplying by the existing coefficient 3, leaving an incomplete answer of $2 \tan x \sec^2 x$.

BOX 3: REAL DOMAIN & ASYMPTOTES

- **Domain of Definition:** The tangent and secant functions are undefined where $\cos x = 0$. Thus:

$$\text{Domain} = \mathbb{R} \setminus \left\{ x \in \mathbb{R} : x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z} \right\}$$

- **Vertical Asymptotes:** At $x = \frac{\pi}{2} + k\pi$, $y \rightarrow +\infty$ and $\frac{dy}{dx} \rightarrow \pm\infty$.
- **Stationary Points:** $\frac{dy}{dx} = 0 \iff \tan x = 0 \iff x = k\pi$. At these points, $y = 0$ (global minimum).

BOX 4: LEIBNIZ SUBSTITUTION (ALTERNATIVE VERIFICATION)

- Let $u = \tan x \implies y = 3u^2$.
- $\frac{dy}{du} = 6u$.
- $\frac{du}{dx} = \sec^2 x$.
- By Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 6u \cdot \sec^2 x = 6 \tan x \sec^2 x$$

BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 3 / 4 MARKS)

- **1.0 Mark:** Correct statement and application of power rule on $\tan x$.
- **1.0 Mark:** Correct inner derivative $\frac{d}{dx}(\tan x) = \sec^2 x$.
- **1.0 Mark:** Multiplying constants $3 \times 2 = 6$ and stating $6 \tan x \sec^2 x$.